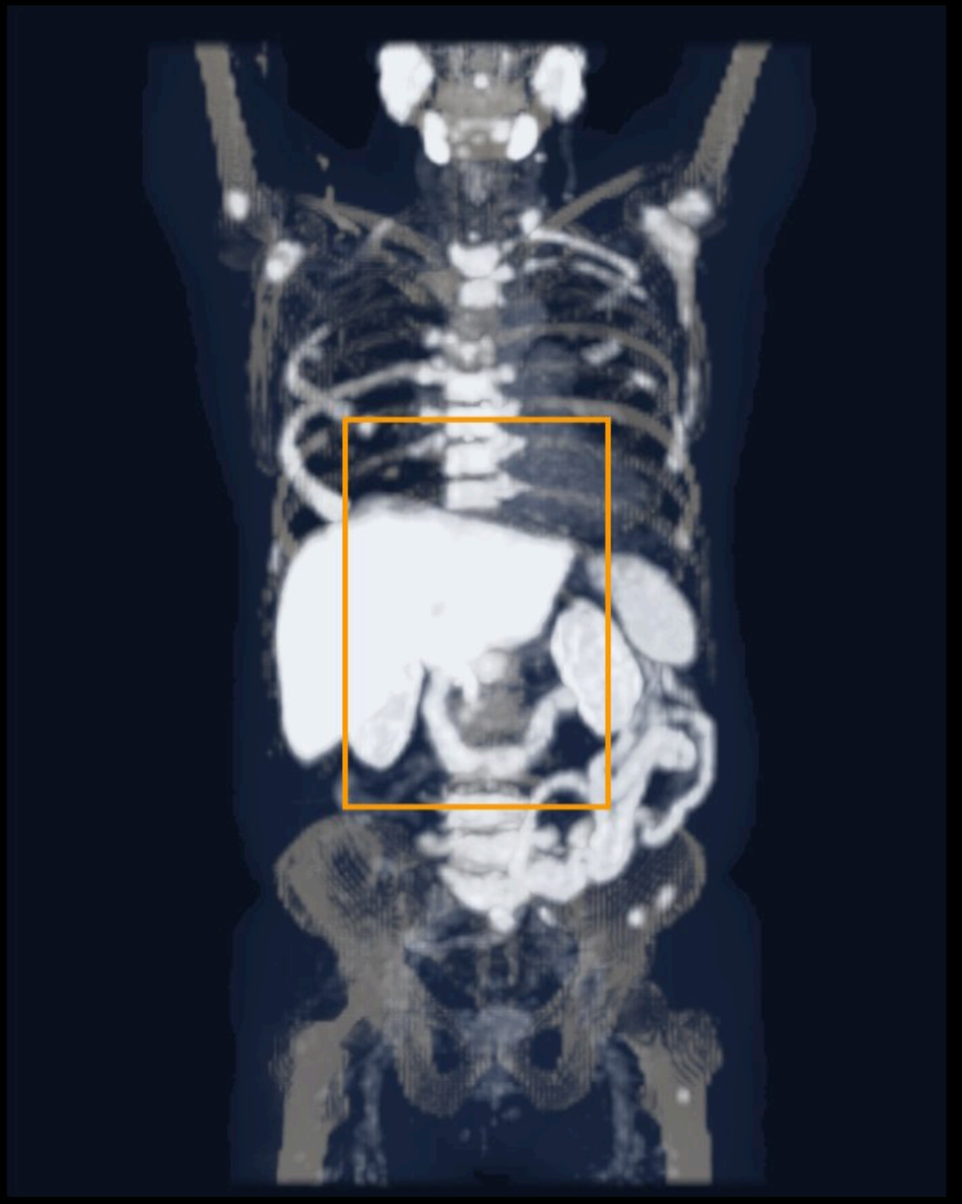
How much of the body does a model need to see?

Segmenting Metastatic Prostate Cancer in Whole-Body PET Scans with a Hybrid CNN-Mamba Network

James Wigfield

Supervised by Dr Jake Kendrick and Dr Mubashar Hassan

THE UNIVERSITY OF WESTERN AUSTRALIA | MEDICAL PHYSICS RESEARCH GROUP



Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)



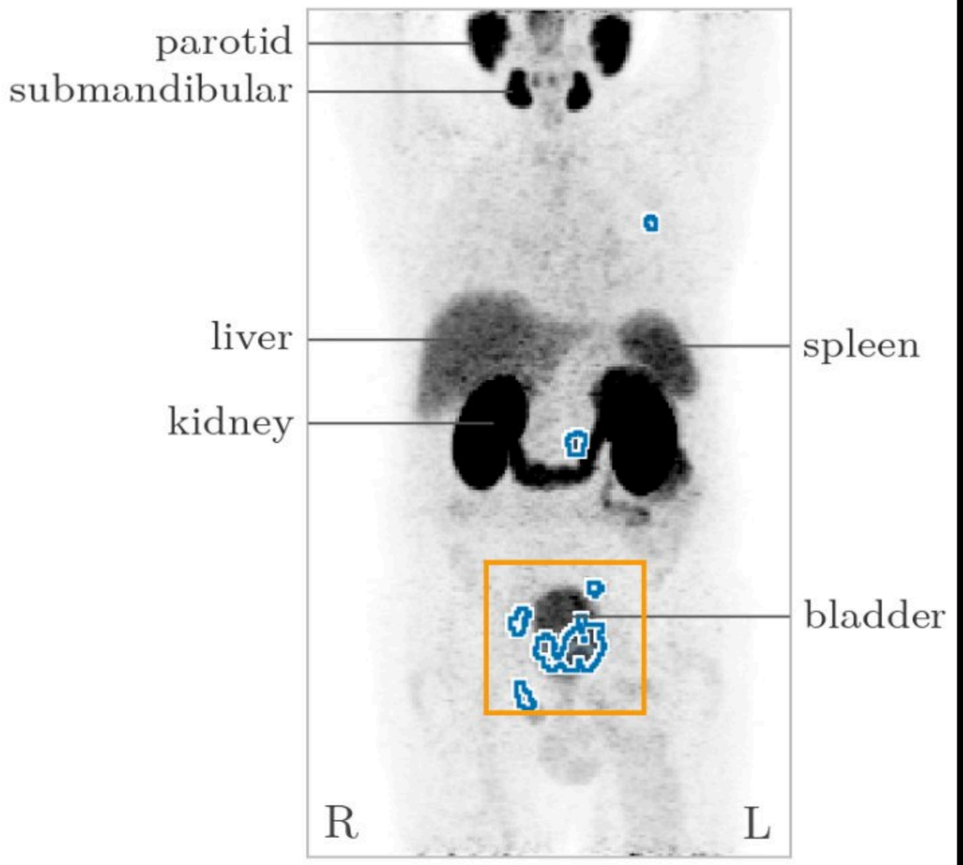
Outlining every lesion by hand is slow, and experts disagree

THE TALK IN ONE BREATH

- Small, scattered lesions; healthy organs glow too
- Mamba could buy more context
- Mamba in the deepest three stages, plus an organ task
- 697 scans, 130 held out
- 82% found at 86% precision, then a twist

[1] Kendrick et al., Eur J Nucl Med Mol Imaging 2022
Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

Healthy organs glow too, so a model needs context



ground-truth lesion mask

200 mm

Up close: a hot spot. Is it cancer?

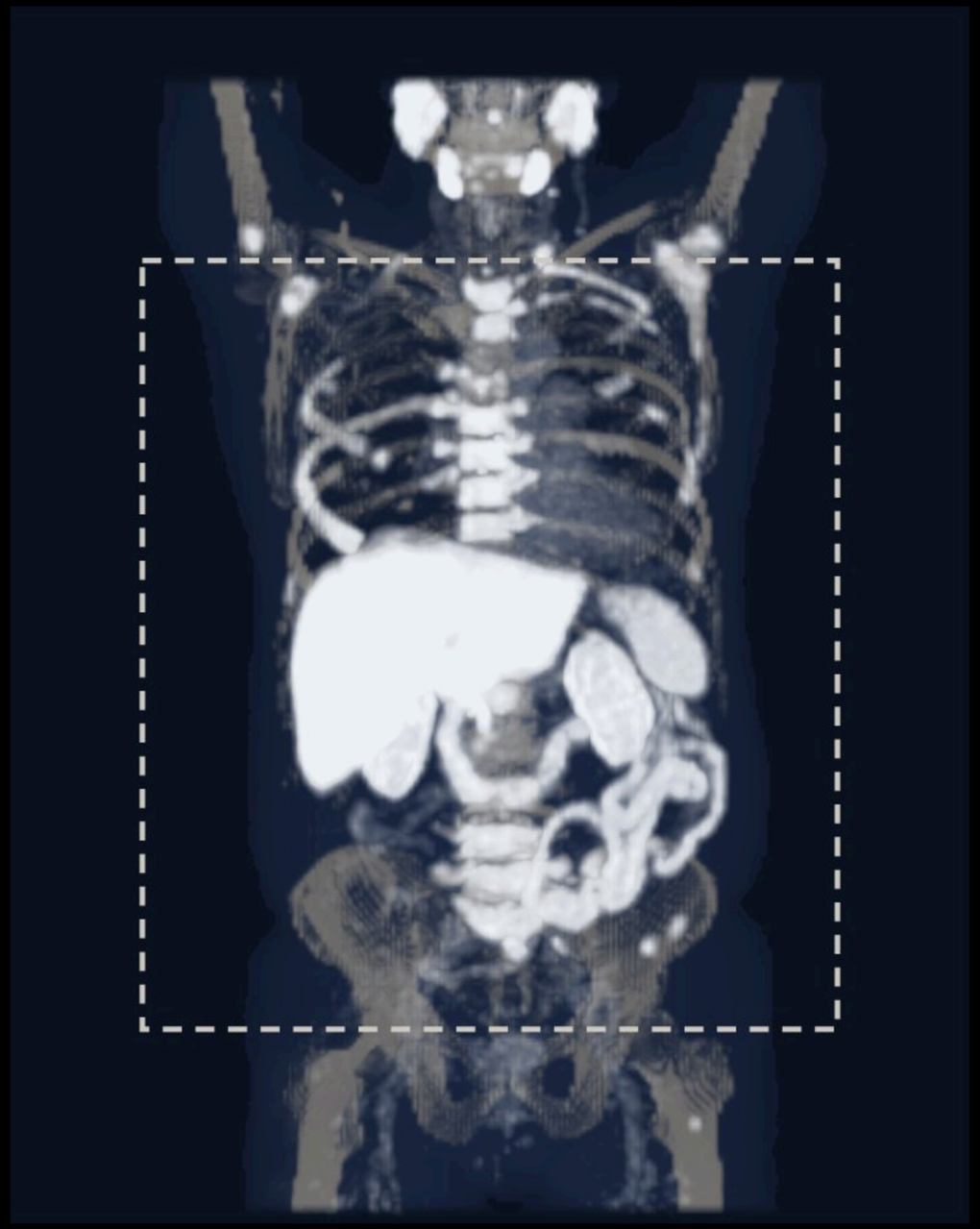
Pulled back: the bladder, with lesions beside it. Glands, liver, spleen and kidneys take up the tracer too.

Telling them apart needs the anatomy around each spot.

Dissertation Figure 2.1 (a): an autoPET-III study, expert-corrected outlines in blue.

[2] Sheikhbahaei et al., Eur J Nucl Med Mol Imaging 2017 · [3] Rokuss et al., arXiv 2024

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)



A CNN reads the body one window at a time

nnU-Net, the standard tool, is a CNN that configures itself to each dataset.

---- **PSMASegmentator's window**
521 mm x 576 mm

Left-right x head-toe, to scale on case A. Depth not shown.

GPU memory caps the window: bigger window, smaller batch.

CONTEXT ONLY · ITS OWN TEST SET

2022 nnU-Net cascade (Kendrick et al.)

Precision 88.2% · found 73.0% · F1 79.9%

128 recurrent-disease test scans, its own matching rule

[4] Isensee et al., Nature Methods 2021 · [1] Kendrick et al., Eur J Nucl Med Mol Imaging 2022
Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

Mamba's linear cost could buy a bigger window

RESEARCH QUESTION

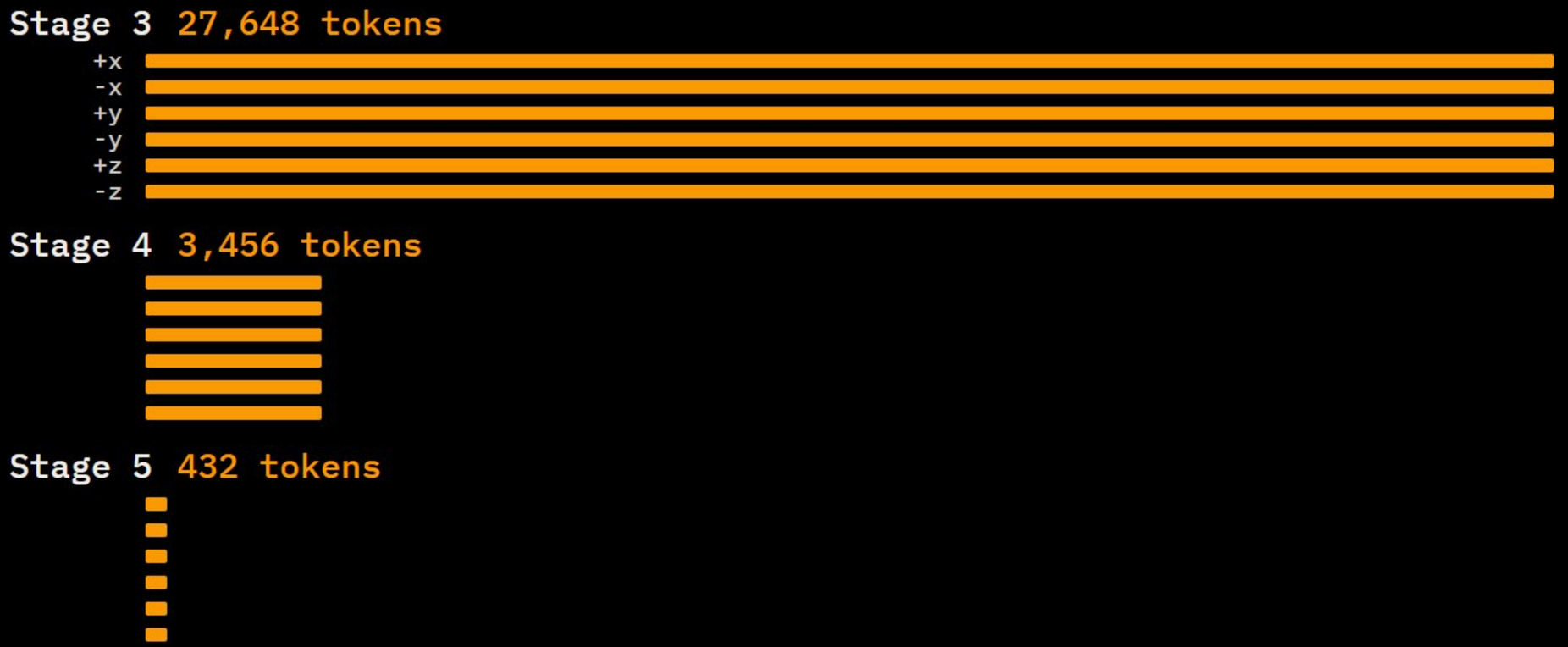
Does a hybrid CNN-Mamba network find lesions better than PSMASegmentator on scans neither model trained on, and do its Mamba blocks add anything over a CNN-only twin?

H1	**Detection**	The hybrid beats PSMASegmentator in pooled F1 on 100 scans neither model trained on.	◌ pending
H2	**Loss**	A loss weighted against misses lifts sensitivity and keeps precision high.	◌ pending
H3	**Efficiency**	At equal network passes, the hybrid is faster and uses less memory.	◌ pending
H4	**Context**	A bigger training window helps the hybrid more than its CNN-only twin.	◌ pending

[5] Gu and Dao, arXiv 2023 · [6] Isensee et al., MICCAI 2024

Mamba sits only in the deepest stages, where sequences are short

One Mamba block scans six directions. Lengths to scale.

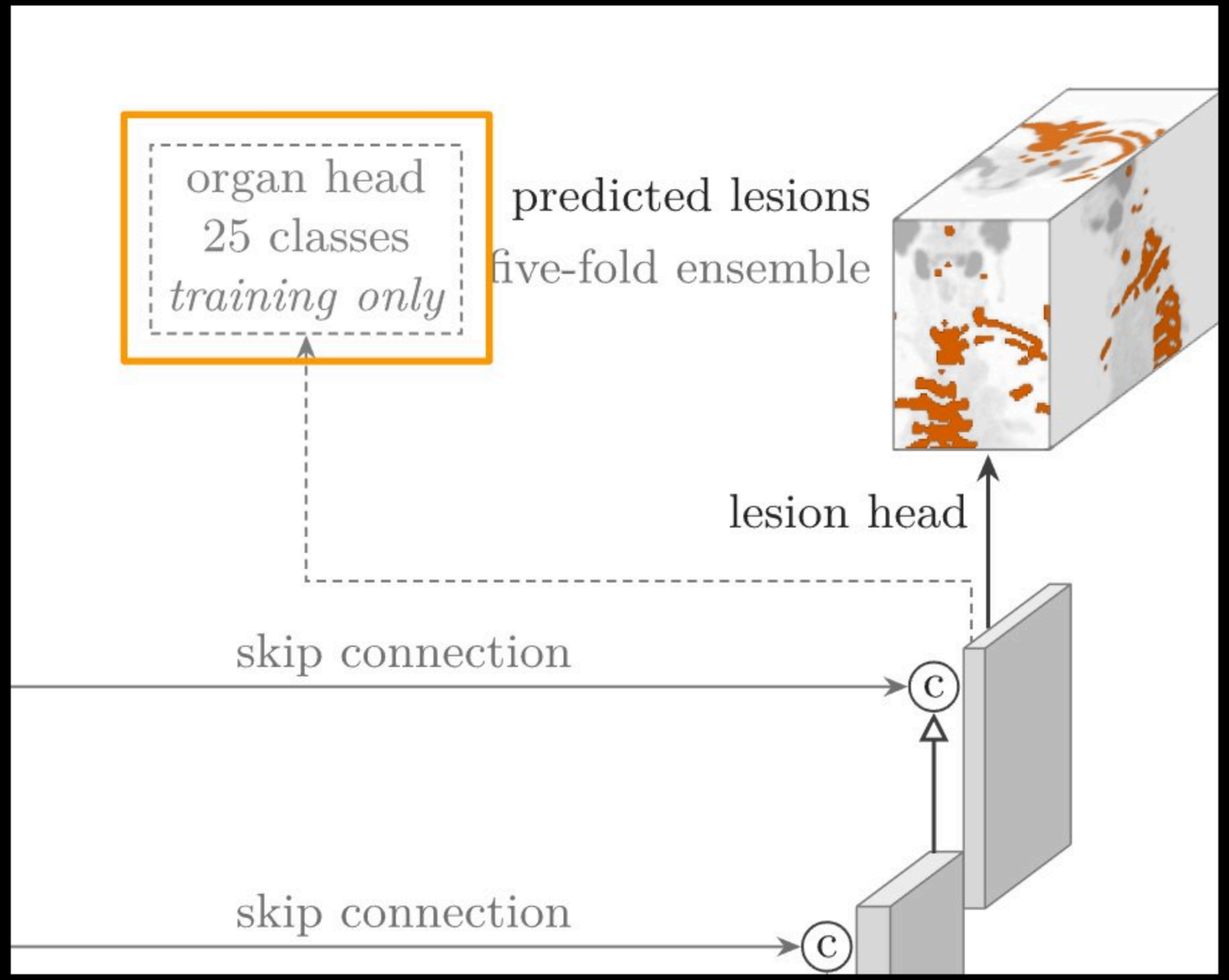


Schematic: one lane per scan direction, one token per feature-map voxel.

[7] Xing et al., MICCAI 2024

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

A training-only organ task cut false alarms on lesion-free scans



Organ head: 25 classes, background and 24 anatomical structures, from TotalSegmentator CT labels. Training only.

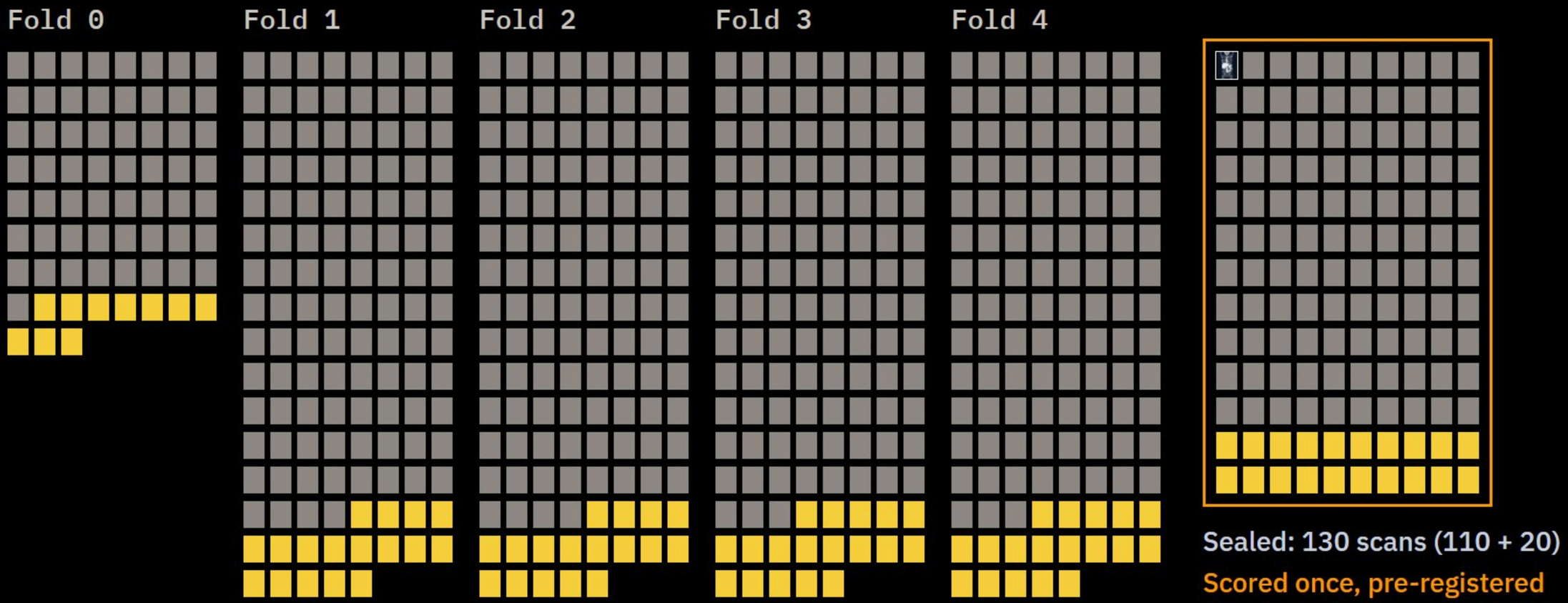
False lesions on the seven lesion-free validation scans

Run 3	**21**
Run 4, with the organ head	**7**

Only the organ head changed between the two runs.

[8] Wasserthal et al., Radiol Artif Intell 2023 · [3] Rokuss et al., arXiv 2024
Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

All 697 scans, with 130 sealed away until the end



[9] Jeblick et al., Sci Data 2026 · [10] Jackson et al., DEEP-PSMA design document 2025

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)



Scored once on the sealed scans, the ensemble found most lesions

84.3%

F1 over all 130 sealed scans, lesions pooled, in-domain

82.4% sensitivity: share of lesions found

86.2% precision: share of what it flagged that was real

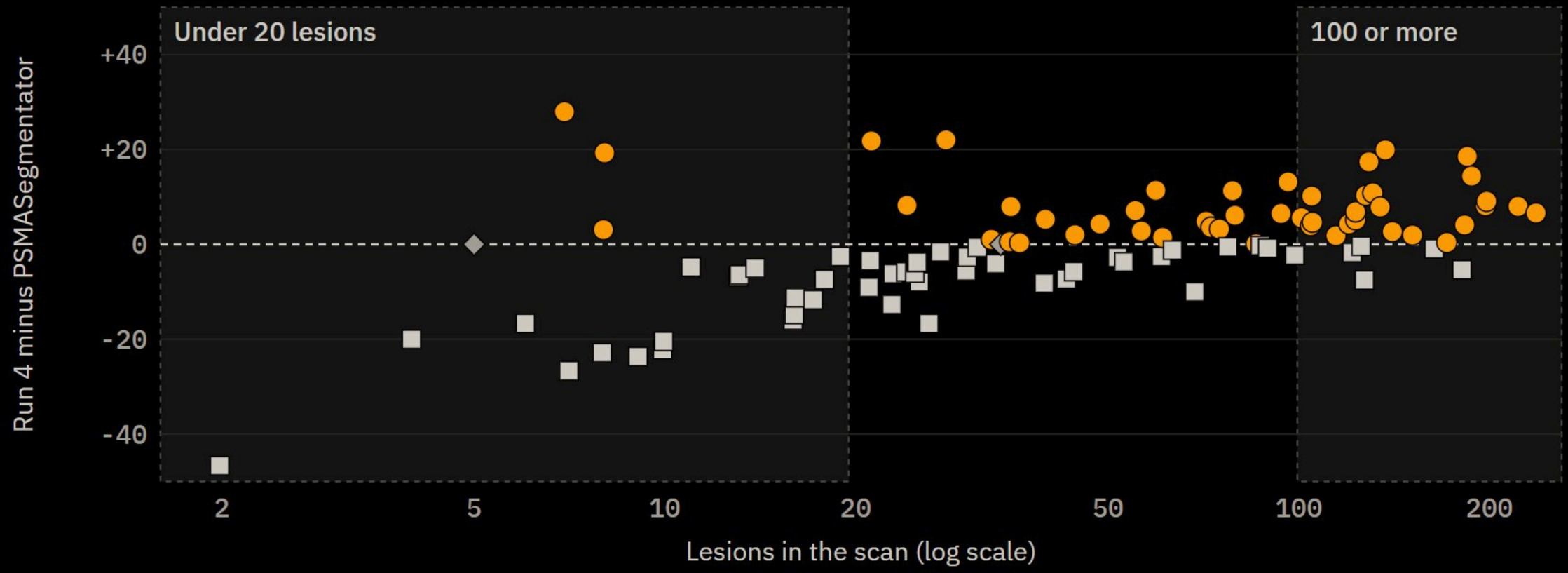
found missed false alarm

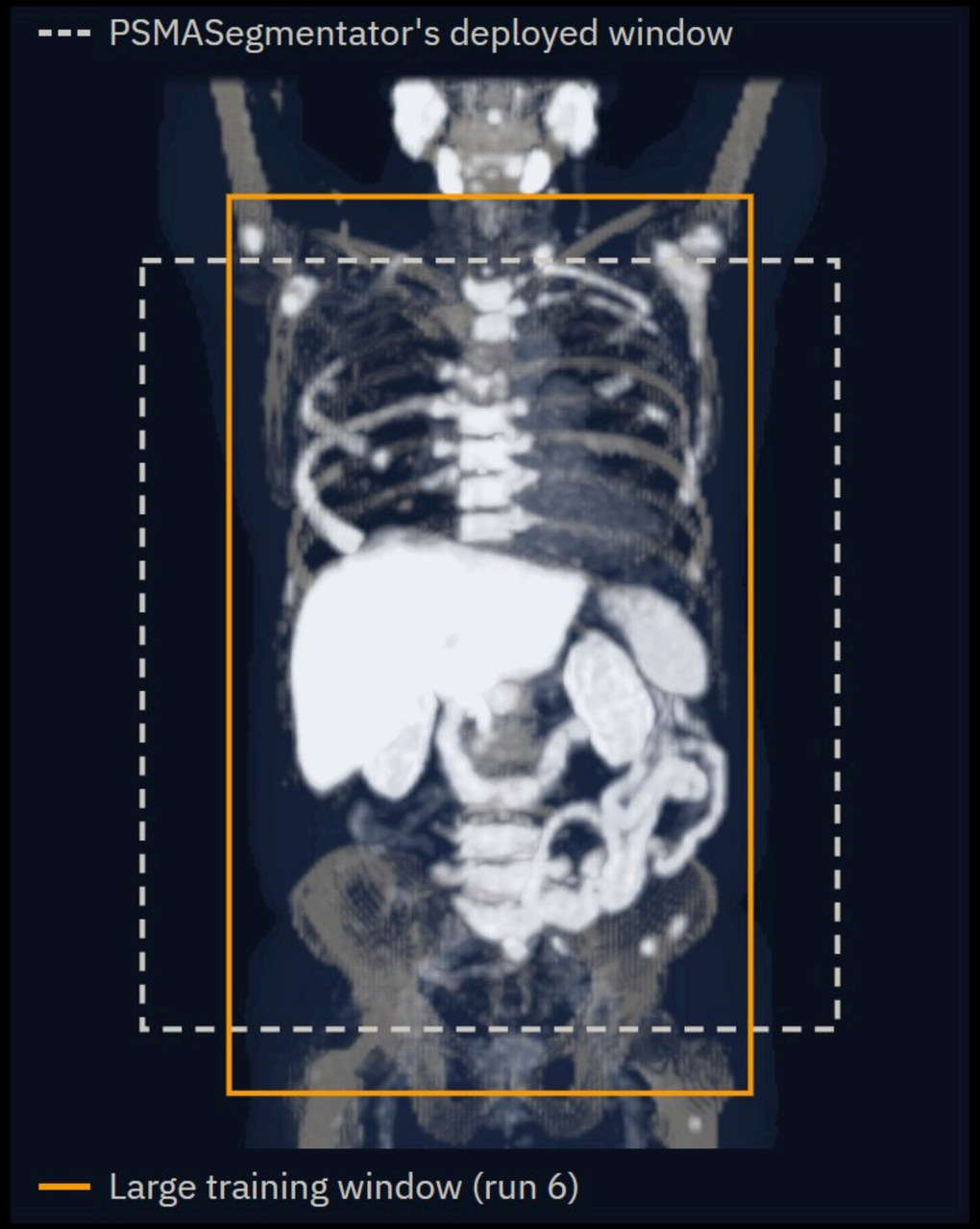
Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

On 100 scans neither saw, run 4 found more lesions than PSMASegmentator

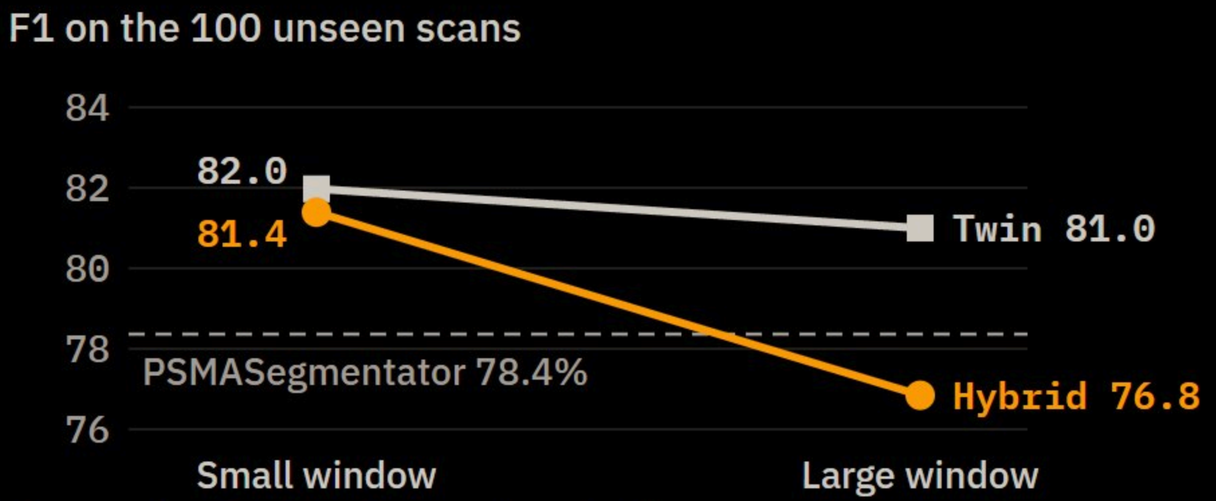
Scan by scan: run 4 better on 48, worse on 50, tied on 2

Pooled F1 +3.0 (1.2 to 4.7)





A bigger window cost the hybrid more F1 than its twin



Interaction -3.57 points: real under the pre-registered rule

Speed at equal passes: no faster in mean time, no lower in peak memory.

The abstract's 32 against 63 s, and 18.1 s against 43.8 s in the dissertation, both come from PSMASegmentator's eight-pass test-time augmentation.

Only run 4's gap to PSMASegmentator was tested with an interval.

[6] Isensee et al., MICCAI 2024

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

As trained, a bigger window helped neither; the gain is not Mamba's

VERDICTS

H1	Detection	◐ supported, as a trade
H2	Loss	✕ not supported
H3	Efficiency	✕ not supported
H4	Context	⇐ reversed on the primary set

A fair test on unseen scans; a pre-registered test of what Mamba adds

LIMITS

- One run per run 6 cell
- Two label conventions
- No SCGH test, no clinical validation

Next: retrain on the SCGH cohort

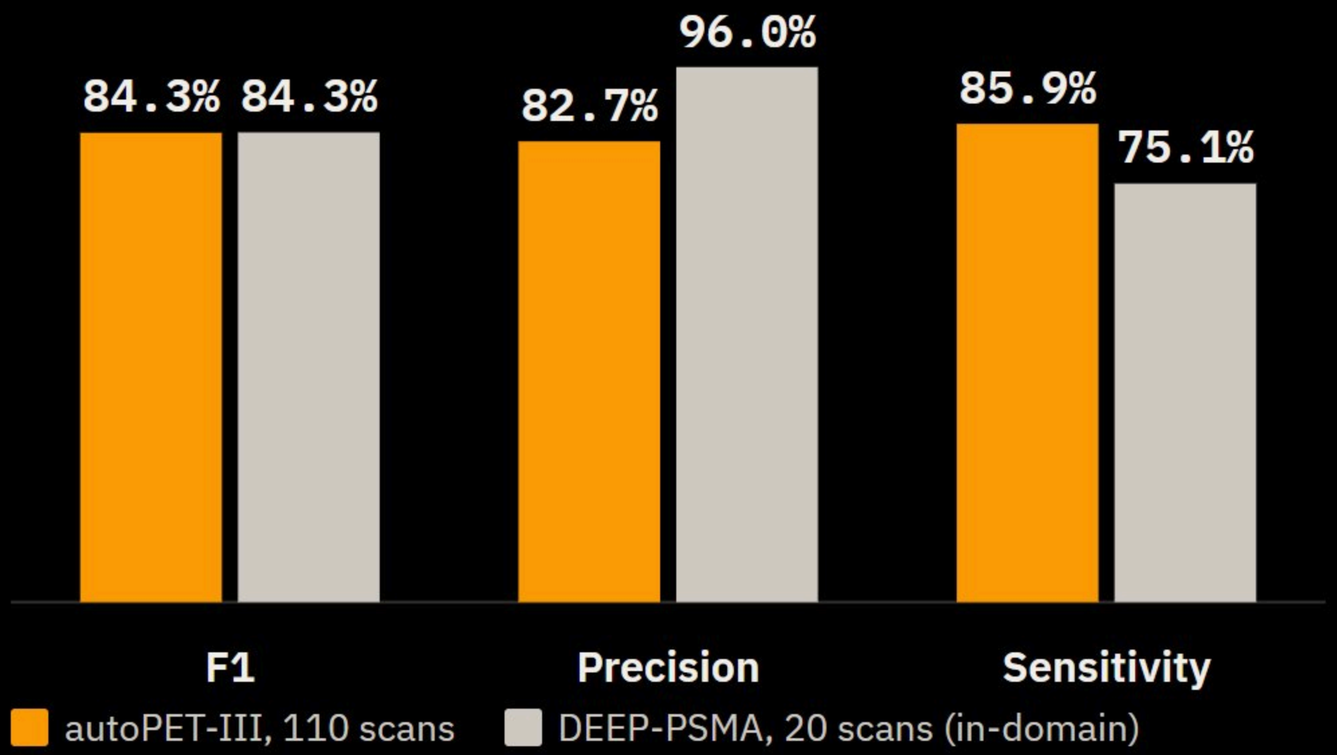
Thank you.

Questions?

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Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

Both cohorts reach F1 84.3%, by different routes



- autoPET-III: expert-corrected outlines
- DEEP-PSMA: every voxel at SUV 3 or more, normal uptake painted out
- About 68 pieces per DEEP-PSMA scan

Cross-validation agreed with the sealed test

81.8%

Cross-validation, 567 scans pooled (fold sd 1.5)

84.3%

Sealed test, 130 scans (81.4 to 86.5)

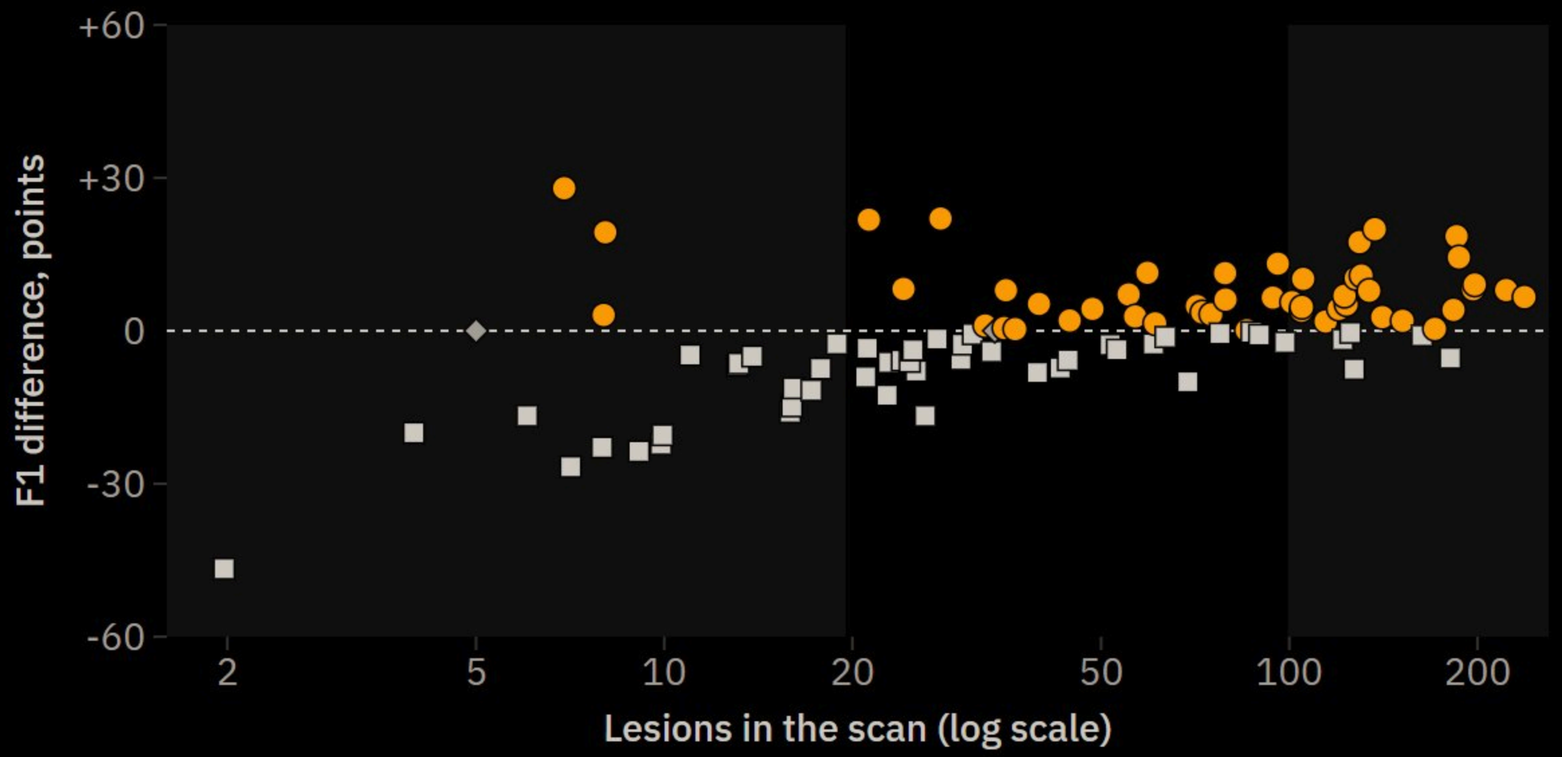
83.1

One fold model alone, plus or minus 0.6

Most recipe differences favour PSMASegmentator

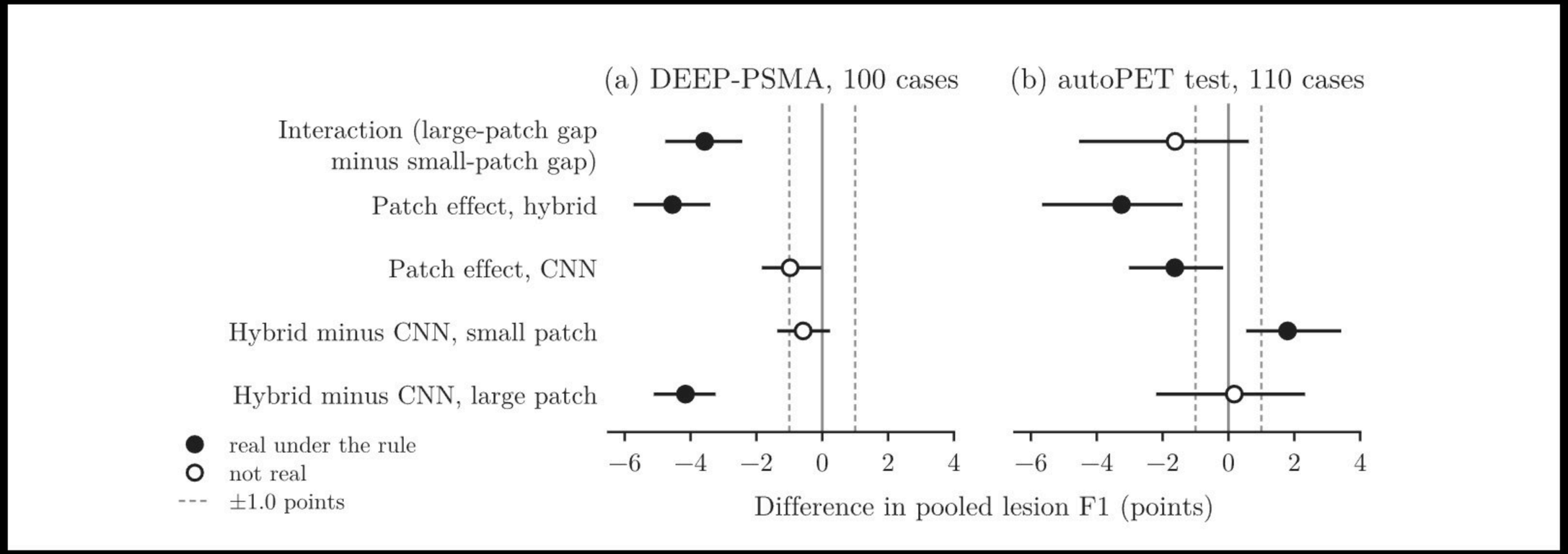
Factor	Run 4 (mine)	PSMASegmentator	Favours
Training scans	430	1,035	PSMASegmentator
Models at inference	one	five averaged	PSMASegmentator
Test-time augmentation	off	eight mirrored passes	PSMASegmentator
Training window	1.77M voxels	4.72M voxels	PSMASegmentator
Parameters	39.7M	102.4M	PSMASegmentator
Optimiser	AdamW	SGD	unclear
Post-processing	none	off for this test	neither
Matching rule, grid, labels	same	same	neither

Scan by scan, there is no clear difference



Run 4 minus PSMASegmentator, per scan. Better on 48, worse on 50, tied on 2. Mean -0.6 (-2.8 to 1.6).

Every run 6 comparison, against the pre-registered rule



Real: interval excludes zero and size at least 1.0 point. One run per cell; large window at batch size 1; large hybrid selected at epoch 275.

The speed gap is test-time augmentation, not the architecture

Configuration	Passes per tile	Mean per scan
Hybrid, as scored	5	18.1 s
PSMASegmentator, default	40	43.8 s
PSMASegmentator, augmentation off	5	17.2 s
Hybrid, augmentation on	40	131.2 s

Development runs 1 to 4 fixed stability, resolution and false alarms

Run	What changed	What it showed
1	recall-weighted loss, no gradient clipping	loss spiked to 4,165; weights later lost
2	clipping, tiled validation (same loss)	precision about 61% (old matcher)
3	2 mm grid, Dice loss, bigger window	false alarms on all seven lesion-free scans
4	organ head	same predictions: F1 73.7 under the old matcher, 85.5 under any-overlap

Mamba sits only where the flattened sequence is short

Stage	Channels	Tokens per direction	Block
1	32	1,769,472	convolution
2	64	221,184	convolution
3	128	27,648	convolution + 2 Mamba
4	256	3,456	convolution + 2 Mamba
5	320	432	convolution + 4 Mamba

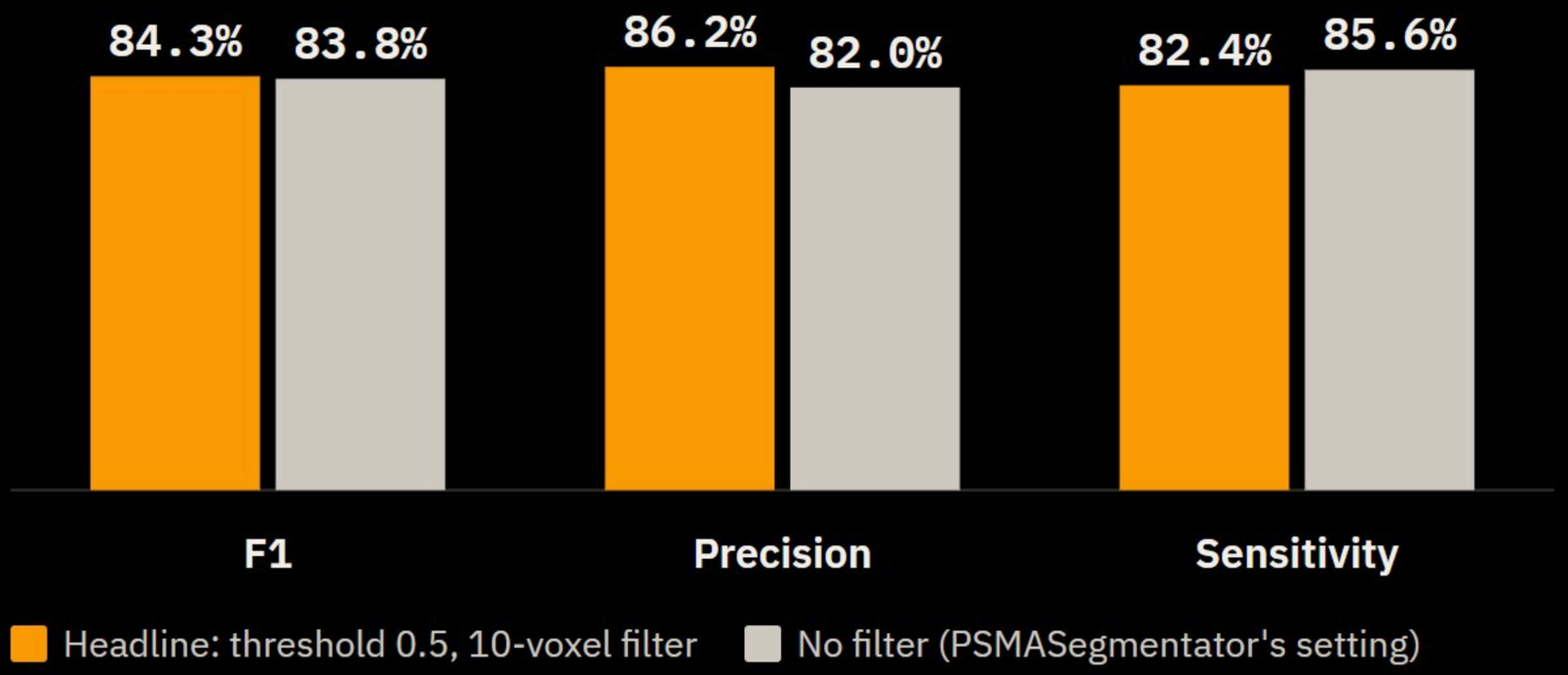
Case B: both lesions found, 14 false alarms



- 8 false alarms in bone, 6 in soft tissue
- None in the kidneys or bladder
- Not the worst scan: nine had no lesion found
- Chosen by rule before viewing

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

The threshold and size filter were fixed before the test



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[1] J. Kendrick, R. J. Francis, G. M. Hassan, P. Rowshanfarzad, J. S. L. Ong, and M. A. Ebert, "Fully automatic prognostic biomarker extraction from metastatic prostate lesion segmentations in whole-body [68Ga]Ga-PSMA-11 PET/CT images," European Journal of Nuclear Medicine and Molecular Imaging, vol. 50, pp. 67-79, 2022.

[2] S. Sheikhbahaei, A. Afshar-Oromieh, M. Eiber et al., "Pearls and pitfalls in clinical interpretation of prostate-specific membrane antigen (PSMA)-targeted PET imaging," European Journal of Nuclear Medicine and Molecular Imaging, vol. 44, pp. 2117-2136, 2017.

[3] M. Rokuss, B. Kovacs, Y. Kirchhoff, S. Xiao, C. Ulrich, K. H. Maier-Hein, and F. Isensee, "From FDG to PSMA: A hitchhiker's guide to multitracer, multicenter lesion segmentation in PET/CT imaging," arXiv preprint arXiv:2409.09478, 2024.

[4] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation," Nature Methods, vol. 18, no. 2, pp. 203-211, 2021.

[5] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," arXiv preprint arXiv:2312.00752, 2023.

[6] F. Isensee, T. Wald, C. Ulrich, M. Baumgartner, S. Roy, K. Maier-Hein, and P. F. Jaeger, "nnU-Net revisited: A call for rigorous validation in 3D medical image segmentation," in Medical Image Computing and Computer Assisted Intervention (MICCAI 2024), 2024.

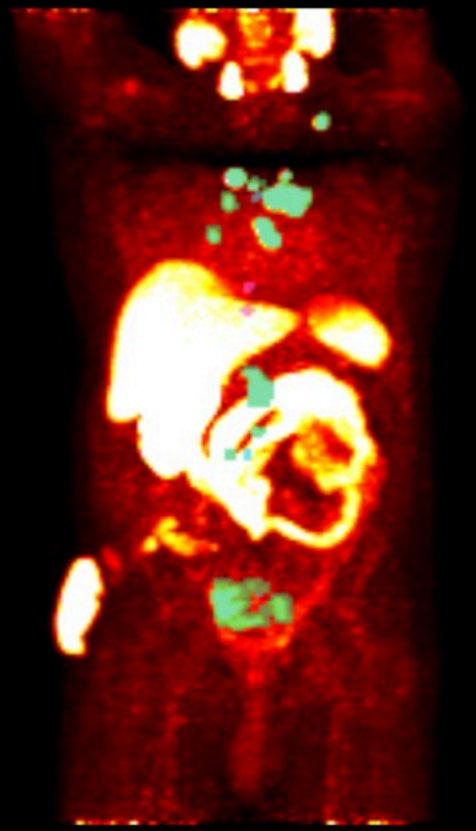
[7] Z. Xing, T. Ye, Y. Yang, G. Liu, and L. Zhu, "SegMamba: Long-range sequential modeling Mamba for 3D medical image segmentation," in Medical Image Computing and Computer Assisted Intervention (MICCAI 2024), 2024.

[8] J. Wasserthal, H.-C. Breit, M. T. Meyer, M. Pradella, D. Hinck, A. W. Sauter, T. Heye, D. T. Boll, J. Cyriac, S. Yang, M. Bach, and M. Segeroth, "TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images," Radiology: Artificial Intelligence, vol. 5, no. 5, p. e230024, 2023.

[9] K. Jeblick, B. Schachtner, A. Mittermeier et al., "A whole-body PSMA-PET/CT dataset with manually annotated tumor lesions," Scientific Data, vol. 13, p. 1023, 2026.

[10] P. Jackson, L. McIntosh, J. Buteau, and M. Hofman, "Deep-learning evaluation for enhanced prognostics - prostate specific membrane antigen (DEEP-PSMA)," Challenge design document (Biomedical Image Analysis Challenges initiative), Zenodo, 2025, not peer-reviewed.

The full recorded run in my app



The same run, turned in 3D

- Found
- Missed
- False alarm

The app's own colours. Five-model ensemble, held-out test scan.

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)